Speaker Verification in Noisy Environment Using GMM Supervectors

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***Abstract*—This paper explores the GMM-SVM combined approach for Text-Independent speaker verification in noisy environment. In recent years supervectors constructed by stacking the means of adapted Gaussian Mixture Models (GMMs) have been used successfully for deriving sequence kernels. Support Vector Machines (SVMs) trained using such kernels provide further improvement in classification accuracy. Analysis of the behavior of such hybrid systems towards simulated noisy data is the object of our study. In our work we have used the KL-divergence and GMM-UBM mean interval kernels for SVM training. All experiments are conducted on NIST-SRE-2003 database with training and test utterances degraded by noises (car, factory & pink) collected from the NOISEX-92 database, at 5dB & 10dB SNRs. A significant improvement of performance is observed in comparison to the traditional GMM-UBM based system.**

***Index Terms*—Speaker Verification, Gaussian Mixture Models, Support Vector Machines, Supervectors, Universal Background Model, Kernel.**

I. INTRODUCTION

Speaker Verification (SV) is the task of validating the claimed identity of a person from his/her voice. It is a binary classification problem in which we distinguish between a true speaker and an impostor. Acoustic modeling is used in this process to effectively capture the distribution of features unique to a speaker. The traditional GMM-UBM based speaker verification system requires a Universal Background Model (UBM) [1] and an adapted Gaussian Mixture Model (GMM) to represent the impostor and actual speaker classes, respectively. During the evaluation stage, a test utterance is classified based on its statistical similarities with the claimed target speaker model (GMM) and the background model (UBM). Peformance accuracy of a SV system is usually increased when these generative models are brought into a discriminative framework using Support Vector Machines (SVM) [2].

Supervectors constructed by stacking the means of an adapted GMM are used to represent speaker characteristics conveyed by speech utterances. Conventionally, a GMM based system calculates log-likelihood probabilities (scores) of features extracted at a frame level. In contrast, supervectors provide numerical comparison of speech utterances as an entire sequence rather than frame-wise probabilities [3]. Relevant speaker information that are often discarded at the frame level has been shown to be preserved in long-term speech utterances [4]. Due to its high dimensionality a supervector makes linear classification much more effective. Kernel functions derived from supervectors which provide a measure of dissimilarity between the target and non-target speaker distributions can thus be an efficient discriminator when used in conjunction with a support vector machine. Another distinct advantage of using supervectors is their ability to compensate for speaker and channel variability through eigenvoice and eigenchannel methods [5].

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The performance of a SV system is considerably reduced in noisy environment [6]. Since accurate estimation of the nature of noise is infeasible, traditional methods aim to compensate for noise degradation. Compensation can be done in all stages of building a SV system i.e., signal processing, feature extraction, acoustic modeling and score calculation. Two popular techniques for compensation at the model-level are maximum *aposteriori* adaptation (MAP) [1] and maximum likelihood linear regression (MLLR) [7]. These methods work by adapting the learned model parameters using speech utterances in testing conditions. MAP is usually used to adapt GMM parameters, when sufficient enrollment data is available. In our present work we try to demonstrate the performance of a SV system in noisy environment using the GMM supervector approach. The work is motivated in part by the diverse applications of SV systems in hand-held devices. Robustness towards noise degradation is desired in any such application. We have used the KL divergence kernel [8] and GMM-UBM mean interval (GUMI) kernel [9] for SVM training. We compare our results with the standard GMM-UBM approach.

The rest of the paper is organized as follows. Section II discusses the GMM-SVM hybrid approach for speaker verification. The experiments carried out are discussed in Section III, results and discussion in Section IV and a brief conclusion of our work in Section V.

II. GMM-SVM APPROACH FOR SPEAKER VERIFICATION

In this section we describe the process of combining the standard GMM based SV system with the SVM framework. This includes a brief intoduction into the GMM-UBM technique followed by supervector construction, kernel design and SVM training.

A. GMM Supervector Construction

The UBM is a GMM trained using a large amount of speech data recorded from multiple speakers over various

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channels [1]. It represents the speaker independent distribution of features. The GMM is built using the Expectation Maximization (EM) algorithm as discussed in [10]. A large number of Gaussian components (typically 1024) are used to avoid overfitting [11]. The standard speaker verification task can be formulated as a hypothesis testing problem in which given a speech segment, the task is to decide whether it is generated from a hypothesized speaker model (null hypothesis) or the background model (alternative hypothesis). The hypothesized speaker models (GMMs) are obtained by modifying the pre-estimated parameters (mean, covariance and weight) of each UBM component using the speaker's training data by Maximum *aPosteriori* (MAP) adaption [1]. A supervector for each speaker model is obtained as follows:-

i. Fixed length vectors for each GMM component is obtained by a kernel mapping function (as discussed in Section II-B). The vector corresponding to the ith GMM component constitutes the ith element of the supervector (subvector).
ii. All the subvectors are concatenated to obtain a high-dimensional vector called the Supervector.

B. KL-divergence and GMM-UBM Mean Interval kernels

The Kullback Leibler (KL) divergence is a non-symmetric distance measure between two probability distributions. Given two distributions p_a and p_b, the KL divergence is defined as

$$D_{KL}(p_a||p_b) = \int p_a(x) log\left(\frac{p_a(x)}{p_b(x)}\right) dx$$

However the KL divergence doesn't satisfy the Mercer's condition for a valid kernel. As a solution a symmterized version of the KL divergence, obtained by bounding the expression by log-sum inequality was proposed as a kernel in [8]. The final version was a linear function of two MAP adapted GMMs p_a and p_b corresponding to utterances a and b. Ignoring adaptation of the UBM covariance matrix Σ_i^u and weights w_i, the resulting Kernel is given by

$$K_{KL}(p_a||p_b) = \sum_{i=1}^{M} (\sqrt{w_i}(\Sigma_i^u)^{-1/2} m_i^a)^T (\sqrt{w_i}(\Sigma_i^u)^{-1/2} m_i^b)$$

where m_i^a and m_i^b are the ith component means of p_a and p_b respectively. Thus the ith subvector of the GMM supervector for any utterance λ is given by

$$s_i^\lambda = \sqrt{w_i}(\Sigma_i^u)^{-1/2} m_i^\lambda \quad i = 1, 2, ..M$$

The Bhattacharya distance between two probalility distribution p_a and p_b is given by

$$D_{Bhatt}(p_a||p_b) = \int \sqrt{p_a(x) p_b(x)} dx$$

For multivariate Gaussian distributions, computing this measure requires estimation of the covariance matrices which in turn demands a high amount of training data. Hence this measure is avoided in practical scenarios. However, it was shown in [12] that second order statistics derived from limited amount of training data could contain supplementary information, when used effectively. The GMM-UBM Mean Interval (GUMI) kernel based on the Bhattacharya distance between GMMs p_a and p_b, as proposed in [12] is given by

$$K_{GUMI}(p_a||p_b) = \sum_{i=1}^{M} (m_i^b - m_i^a)^T \left[\frac{\Sigma_i^a + \Sigma_i^b}{2}\right]^{-1} (m_i^b - m_i^a)$$

Considering the statistical similarities of a adapted speaker GMM and the UBM the ith subvector of the GMM supervector for any utterance λ is given by

$$s_i^\lambda = \left[\frac{\Sigma_i^\lambda + \Sigma_i^u}{2}\right]^{-1/2} (m_i^\lambda - m_i^u) \quad i = 1, 2, ..M$$

For any utterance the final supervector obtained by concatenating the subvectors is given by $S = [s_1^T, s_2^T, ...s_M^T]^T$.

C. SVM training

A support vector machine is a binary classifier [11]. Using labeled training vectors, a SVM optimizer finds a decision hyperplane that maximizes the margin of separation between two classes (target speaker and impostor). The classifier equation is given as follows:-

$$y(x) = \sum_{i=1}^{L} \alpha_i t_i K(x_i, x) + d$$

Where $\alpha_i > 0$ are the Lagrange multipliers and x_i are the Support Vectors. Both these parameters are learned during the optimization process. $t_l \in$ {-1,+1} are the ideal outputs, K is the desired kernel mapping and d is a bias parameter. For any vector x the actual output $y(x)$ is compared with a decision threshold for final classification.

III. PROPOSED STUDIES

All experiments were conducted on the NIST-SRE-2003 database [13] which consists of 356 target speaker utterances (149 male, 207 female). Around 75 secs of speech from each target utterance was used for training and 3500 utterances of roughly 15 secs each for testing. In the next two subsections we describe the SV sytems developed using GMM-UBM and the GMM-SVM approaches, respectively. The GMM-UBM based SV sytem was used as baseline.

A. GMM-UBM approach

1) Simulation of noisy training and test utterance: For a relative reduction of mismatch, both the training and test speech utterances were corrupted by noise. The assumption here is that the nature of noise in the test speech signal and its SNRs are known in advance. Each target speaker utterance was reconstructed using 3 different additive noises i.e., *car noise*, *factory-1 noise* and *pink noise* collected from the NOISEX-92 database [14]. In our work we explored two different SNRs at 5dB and 10dB respectively, for all speech utterances. The test utterances were similarly corrupted by the same noises at the same SNRs. Thus we have 6 sets of target speaker and test utterances, each corrupted by a particular noise at either of the two SNRs. Each such set was used separately for feature extraction, acoustic modeling and performance evaluation.

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2) Feature Extraction & Normalization: All the speech utterances were processed using short term frames of 20 ms with a frame overlap of 10ms. In the preprocessing stage, an energy based thresholding was used for voiced activity detection. The average frame energy of an entire utterance was calculated. Frames with energy less than 1% of the average energy were carefully discarded as silence/unvoiced region. 13-Mel Frequency Cepstral Coefficients (MFCCs) [15] derived from a 26-channel Mel-scaled filter bank were extracted from each frame. The zeroth coefficient was excluded. First and second order delta coefficients computed over a frame span of 2, were appended to the 13 MFCCs giving a resultant 39 dimensional feature vector. For noise compensation all vectors were subjected to cepstral mean subtraction [16]. The resultant vectors were scaled to unit variance by cepstral variance normalization (CVN).

3) Acoustic Models: The SwitchBoard II corpus was used for building the Universal Background Model (UBM). The data consists of six minutes of conversational speech from each of 100 male and 100 female speakers recorded over a cell phone. The male and female speaker's data were pooled separately into two gender-dependent sets each having 10 hours (6 X 100 mins) of speech. Two gender dependent GMMs of 512 components each were estimated from each set using the EM algorithm. A 1024-component gender independent UBM was constructed by pooling the two GMM parameters together and normalizing the weights.

To obtain the hypothesized target speaker models, the UBM parameters were adapted by MAP technique using the noisy target speaker utterances as described in the Section II. For each of the 6 individual sets of noisy training data, we thus have 356 target speaker models (GMMs).

4) Evaluation: The standard evaluation technique for NIST-SRE-2003 was used. Each test utterance was scored against 11 hypothesized target speaker models. The *'false alarm'* and *'miss'* error rates were computed. A weighted sum of these errors was used to determine the Detection Cost Function (DCF). We used the equal error rate (EER) and the minimum DCF value as the metrics for performance evaluation.

B. GMM-SVM approach

1) Acoustic Vector Resampling: In the pre-processing stage of SVM training, we tried to counter a major drawback of the conventional SVM scoring system. Since the number of background speaker utterances strictly outnumber the number of enrolled utterances (typically one), there is a major data imbalance. This makes the SVM decision boundary skew towards the minority class causing high false rejection rate unless the decision threshold is properly set to compensate for the bias. As a solution we used a recently proposed version of minority oversampling technique called acoustic vector resampling (AVR) [17] [18]. The method works as follows:-

i. For each target speaker utterance of the NIST database, MFCC features were extracted.
ii. The sequence of occurrence of feature vectors was randomized and the resultant sequence was divided into 2 partitions (sub-sequences).
iii. Steps 1 & 2 were repeated thrice to obtain a total of 6 sub-sequences.
iv. These 6 sub-sequences together with the original full length sequence was used as enrollment data for a target speaker.
v. The remaining 355 speakers were used as background. In contrast to the target speaker utterances, each background speaker utterance was subjected to AVR twice. This gave us a total of 3 sequences for each background speaker.

Though this method increases the computational load on a support vector machine, the performance as shown in Section IV is comparitively much better.

2) Kernel Design & SVM Training : For each target speaker of the NIST database we obtained 7 enrollment utterances and 1065 background utterances as discussed in Section III-B1. For each utterance, a GMM supervector was constructed and a kernel designed as described in details in Section II. Each supervector consists of 39936 elements (39-dim mean, 1024-component UBM). With the enrollment supervectors labeled +1 and background supervectors labeled -1, a SVM was trained for each target speaker using a specific kernel (KL-divergence, GUMI). The training process was repeated individually for training data in each of the 6 different noisy datasets as discussed in Section III-A1.

3) Evaluation: The 3500 test utterances from each noisy dataset were transformed into supervectors prior to evaluation. Each test utterance was scored against 11 hypothesized speaker models (SVMs) from the same noisy dataset. Given a noisy test utterance X^{test} the Kernel scoring was obtained as follows:-

$$Score(X^{test}) = \sum_{t=1}^{T} \alpha_t K(X^t, X^{test}) - \sum_{i=1}^{B} \alpha_i K(X^i, X^{test}) + d$$

Where X^t and X^i are the sequence of support vectors corresponding to the target and background speaker classes while α_t and α_i are the non-zero Lagrange multipliers of the corresponding SVM models. T and B are the total number of support vectors in each class. d is a bias term and K is either of the two kernels used. The EER and minimum DCF values were used as metrics for performance evaluation.

IV. RESULTS AND DISCUSSION

The performance of (i) GMM-UBM based SV system (baseline) (ii) GMM-SVM based SV system with KL divergence kernel and (iii) GMM-SVM based SV system with GUMI kernel in various noisy environment has been summarized in Table I.

Figure 1 and Figure 2 shows the Detection Error Tradeoff (DET) plots for the GMM-UBM and GMM-SVM systems respectively. Each figure has three curves corresponding to each simulated noisy dataset. In Table I multicolumns 2 and 3 represent the GMM-SVM based systems trained using KL divergence and GUMI kernels respectively. A consistent reduction in the EER values is noticed in case of the SVM based systems across all the noisy backgrounds. It can be further observed that the EER values are comparatively less in case of the systems based on GUMI kernels than the KL

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TABLE I: Performance of the Speaker Verification systems in noisy environments

SNRs	Noises	GMM-UBM		GMM-SVM (KL div)		GMM-SVM (GUMI)	
		EER(%)	MinDCF	EER(%)	MinDCF	EER(%)	MinDCF
5dB	Car	18.11	0.071	7.09	0.032	6.68	0.030
	Factory	20.96	0.085	12.24	0.048	11.74	0.047
	Pink	23.89	0.092	13.19	0.054	11.79	0.046
10dB	Car	15.44	0.068	6.59	0.031	6.23	0.029
	Factory	16.44	0.072	9.44	0.039	8.76	0.038
	Pink	18.65	0.081	10.21	0.043	8.72	0.037

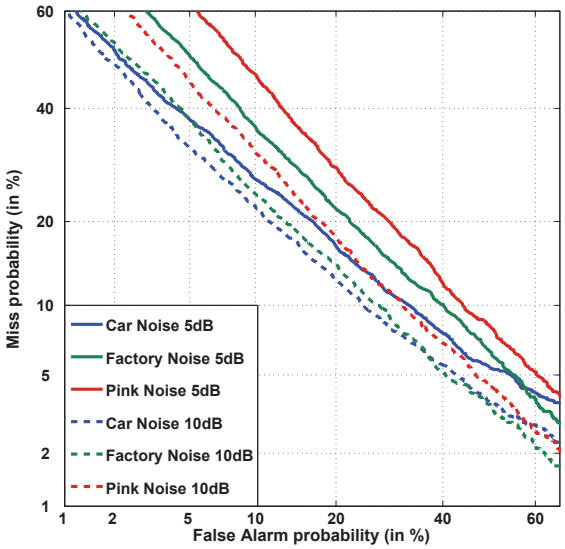
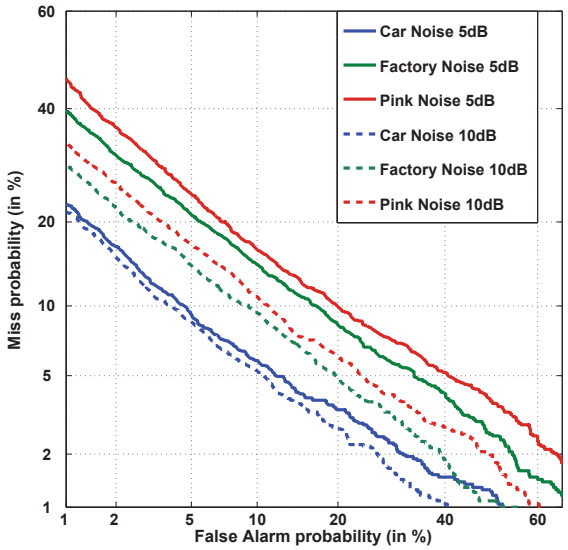
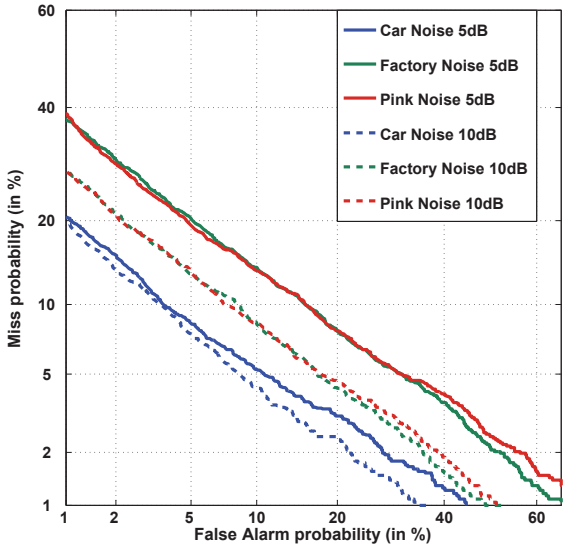


Fig. 1: DET plots for GMM-UBM based SV system in noisy environment



(a) KL divergence kernel



(b) GUMI kernel

Fig. 2: DET plots for GMM-SVM based SV systems in noisy environment with (a) KL divergence kernel (b) GUMI kernel

divergence kernel. The improvement is also valid in case of the minimum DCF values. The relative improvement of the GMM-SVM based systems in terms of EER is shown in Table II. The relative EER is calculated as

$$EER_R = \frac{(EER_B - EER_V)}{EER_B} \; X \; 100\%$$

where EER_B and EER_V are the equal error rates for the baseline and GMM-SVM based systems.

TABLE II: The relative equal error rates for GMM-SVM based SV systems in noisy environments

Noises	Relative equal error rate $EER_R\%$			
	SNR (5 dB)		SNR (10 dB)	
	KL div	GUMI	KL div	GUMI
Car	60.85	63.11	57.32	59.65
Factory	41.60	43.99	42.58	46.72
Pink	44.79	50.65	45.25	53.24

V. CONCLUSION

In this paper we demonstrated the performance of the GMM-SVM hybrid approach for speaker verification on the NIST-SRE-2003 corpus in three simulated noisy environments. The relative improvement in performance accuracy compared to the traditional GMM-UBM based system was shown. It was also observed that the GMM-UBM Mean Interval (GUMI) kernel used for training SVMs performed better than the KL

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divergence kernel for all three noises. In the present work we have explored only 3 different noises at 2 different SNRs (5dB & 10dB). The work can be extended for a larger number of multi-SNR noises in mismatched conditions. Since our focus of study was to analyse the behavior of the system at the model level, we consciously avoided sophistication at the signal or feature level of speech. Performance of the system can be enhanced by processing Vowel-like regions (VLRs) of a speech utterance which are robust towards noise degradation [19]. In addition to spectral features, source and prosodic features may also be explored in future work. The performance can also be further improved by score normalization techniques like T-Norm [20] or channel compensation techniques like Nuisance Attribute Projection (NAP) [21].

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